Nawres Alwan

PhD Candidate in Computer Science — Quantum Cryptography, Quantum Machine Learning and Quantum Information Science

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PROFILE

PhD researcher at Universiti Malaya, Malaysia. Doctoral thesis submitted; viva voce examination expected in December 2026. Formal degree completion and certificate issuance pending. I design quantum representations that preserve the structure of data — spatial, spectral and relational — and study how that structure can make quantum learning and quantum communication more trainable, robust and secure. First author on 11 of 15 published peer-reviewed papers.

Research interests: quantum image representation; quantum machine learning and optimization; quantum cryptography and QKD; chaos and graph theory.

EDUCATION

2021 – Present
PhD in Computer Science (candidate), Universiti Malaya, Malaysia
Faculty of Computer Science and Information Technology. Specialization: Quantum Cryptography and Quantum Computing.
Thesis submitted; viva voce examination expected December 2026. Degree certificate pending.

2012 – 2016
M.Sc., University of Technology, Iraq
Thesis: *Spectral Graph Theory with Applications* — characteristic polynomials and spectral properties of graph classes via adjacency, Laplacian and signless Laplacian matrices; extremal eigenvalues of connected irregular tripartite graphs.

2007 – 2011
B.Sc. in Applied Mathematics, University of Technology, Iraq
Thesis: *Spanning Tree* — methods for computing spanning trees of weighted and unweighted graphs, with a MATLAB program for comparative evaluation.

SELECTED PUBLICATIONS

S1 **Multilayered Quantum Computing and Simulation System for Enhanced Image Representation of HSI Based Fourier Transform and Adjacency Matrix.** Scientific Reports, 2025.
Adjacency-aware quantum image representation for hyperspectral images, combining Fourier information with an adjacency matrix.

S2 **A Multi-Channel Quantum Image Representation Model with Qubit Sequences for Quantum-Inspired Image and Image Retrieval.** AIMS Mathematics, 2025.
Multi-channel encodings (QIRBP, QIRWC) of bit-plane, wavelength and neighbouring-pixel relationships using qubit sequences.

S3 **A Trust Optimization Region of the Quantum Natural Gradient for the Large Damping Domain.** ICCSA 2026 Workshops, Springer.
A trust-region treatment of quantum natural-gradient optimization in the large-damping domain.

S4 **A Comprehensive Review on Classification of Quantum Image Representations.** Archives of Computational Methods in Engineering, 2025.
A classification of quantum image representation models, their characteristics and their limitations.

S5 **Color Image Encryption Through Multi-S-box Generated by Hyperchaotic System and Mixture of Pixel Bits.** Fractals, 2024.
Colour image encryption with multiple S-boxes generated by a hyperchaotic system (classical cryptography).

PUBLICATIONS — QUANTUM COMPUTING AND QUANTUM IMAGE REPRESENTATION

01 **Multilayered Quantum Computing and Simulation System for Enhanced Image Representation of HSI Based Fourier Transform and Adjacency Matrix.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, N. F. Mohd Noor, and Y. Karaca.
Scientific Reports, 2025. doi:10.1038/s41598-025-24168-4 [Journal]

02 **A Multi-Channel Quantum Image Representation Model with Qubit Sequences for Quantum-Inspired Image and Image Retrieval.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, N. F. Mohd Noor, and Y. Karaca.
AIMS Mathematics, 2025. doi:10.3934/math.2025499 [Journal]

03 **A Comprehensive Review on Classification of Quantum Image Representations.**
N. A. Alwan, N. M. G. Al-Saidi, S. J. Obaiys, N. F. Mohd Noor, and M. Ahmad.
Archives of Computational Methods in Engineering, 2025. doi:10.1007/s11831-025-10369-7 [Journal]

04 **A Trust Optimization Region of the Quantum Natural Gradient for the Large Damping Domain.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, and R. Chadli.
Computational Science and Its Applications - ICCSA 2026 Workshops.
Springer, Cham, 4 July 2026. doi:10.1007/978-3-032-30497-1_18 [Conference]

PUBLICATIONS — CRYPTOGRAPHY AND CHAOTIC SYSTEMS

05 **Color Image Encryption Through Multi-S-box Generated by Hyperchaotic System and Mixture of Pixel Bits.**
N. A. Alwan, S. J. Obaiys, N. F. Mohd Noor, N. M. G. Al-Saidi, and Y. Karaca.
Fractals, 2024. doi:10.1142/s0218348x24400395 [Journal]

06 **A Secure Image Cryptosystem via Multiple Chaotic Maps.**
M. T. Wazi, D. S. Ali, N. M. G. Al-Saidi, and **N. A. Alwan**.
Discrete Mathematics, Algorithms and Applications, Vol. 14, No. 4, 2021. doi:10.1142/S179383092150141X [Journal]

07 **A Pseudo Random Number Generator Based on 4D Hyperchaotic Systems, Riddled Basins of Attraction and Advanced Microfluidic Technology.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, N. F. Mohd Noor, and Y. Karaca.
Computational Science and Its Applications - ICCSA 2024 Workshops.
Springer, Cham, 30 July 2024. doi:10.1007/978-3-031-65154-0_6 [Conference]

08 **Secure Image Encryption Using Single-Mode Fiber and Dense Wavelength Division Multiplexing in Chaotic Systems.**
N. M. G. Al-Saidi, S. J. Obaiys, **N. A. Alwan**, A. J. Mohammed, A. K. Farhan, and Y. Karaca.
Springer, 2024. doi:10.1007/978-3-031-65154-0_5 [Conference]

09 **Performance-Enhancing of RSA Public Key via Three-Dimensional Hyperchaotic System.**
N. A. Alwan, A. Y. Yousif, and N. M. Al-Saidi.
AIP Conference Proceedings 2325, 020027, February 2021. doi:10.1063/5.0040397 [Conference]

10 **Image Encryption Based on Highly Sensitive Chaotic System.**
D. S. Ali, **N. A. Alwan**, and N. M. Al-Saidi.
AIP Conference Proceedings 2183, 080007, 2019. doi:10.1063/1.5136200 [Conference]

PUBLICATIONS — SPECTRAL GRAPH THEORY AND DISCRETE MATHEMATICS

11 **Tripartite Graphs with Energy Aggregation.**
N. A. Alwan, N. M. G. Al-Saidi, and R. T. Rasheed.
Boletim da Sociedade Paranaense de Matematica, Vol. 38, No. 7, pp. 149-167, 2019. doi:10.5269/bspm.v38i7.44463 [Journal]

12 **Coding and Descriptions for the Subset of Twin Star Mean Labeling.**
Asia Pacific Journal of Mathematics, 2024. doi:10.28924/APJM/11-59 [Journal]

13 **A General Formula for Characteristic Polynomials of Some Special Graphs.**
N. A. Alwan and N. M. Al-Saidi.
Engineering and Technology Journal, Vol. 34, No. 5B, pp. 638-650, 2016. [Journal]

14 **A New Approach for the Characteristic Polynomial of a Complete Tripartite Graph.**
N. A. Alwan, N. M. Al-Saidi, and W. J. Abdulaa.
AIP Conference Proceedings 2334, 080002, March 2021. doi:10.1063/5.0042627 [Conference]

15 **The Energy of a Tripartite Graph.**
N. A. Alwan, N. M. G. Al-Saidi, and R. T. Rasheed.
AIP Conference Proceedings 2091, 030006, April 2019. doi:10.1063/1.5095091 [Conference]

MANUSCRIPTS UNDER REVIEW (NOT YET PUBLISHED)

P1 **Quantum Encryption Framework via Hyperchaotic Systems and Geometric Phase Randomness for Secure Quantum Images.**
Quantum Information Processing, 2025.

P2 **Geometry-Driven Quantum Randomness: Phase-Sensitivity Amplification and Min-Entropy Certification under Hyperchaotic Control.**
Scientific Reports, 2026.

P3 **A Novel Sector-Conditioned Finite-Key Method for Secure Dynamically Phase-Modulated Quantum Key Distribution Using PSB Geometric Phase.**
Scientific Reports, 2026.

INTELLECTUAL PROPERTY

Patent-related institutional submission, Universiti Malaya — *Relational Quantum Geometry for Image and Spatially Indexed Multidomain Data Processing*. 21 September 2026. Submitted.

Patent-related institutional submission, Universiti Malaya — *Quantum geometric learning system and method for structure-aware image processing based on quantum image representation and Hamiltonian-driven manifold evolution*. 2 September 2026. Submitted.

Copyright institutional submission, Universiti Malaya — *Quantum Deep Learning Strategies for Facial Feature Extraction Based on Quantum Image Representation*. 27 September 2026. Submitted.

FUNDING

Universiti Malaya — Two Graduate Research Assistantship grants in support of doctoral research in quantum cryptography and computational quantum science.

TEACHING EXPERIENCE

Lecturer — **Mathematics, Image Processing and Programming**
Applied mathematics; image processing; MATLAB and scientific programming; computational methods.
Prepared to teach: Python and numerical simulation; quantum computing and algorithms; quantum machine learning; quantum cryptography and QKD.

PROFESSIONAL EXPERIENCE

Jan 2025 – Present — **Graduate Research Assistant**, Universiti Malaya (scheduled end: December 2026)

2024 – Present — **IT Department Manager**, Hezn Al-Aman for Trading and General Contracting

2016 – 2024 — **Director**, Solution Center for Research and Software

SKILLS, CONFERENCES, MEMBERSHIPS AND LANGUAGES

Programming — Python, MATLAB, JavaScript, TypeScript, HTML, CSS, Lean 4

Quantum and scientific — Qiskit, numerical simulation, algorithm development, LaTeX, scientific writing

AI and ML — PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, Tesseract OCR

Web — Django, Next.js

Conferences — ICMS 2018, 2019, 2020 (International Conference of Mathematical Sciences); 3rd Scientific Conference on Computing and Information Technology, Anbar University (2019); ICCSA 2024 and ICCSA 2026 Workshops (Springer).

Memberships — IEEE Member; Iraqi Al-Khwarizmi Association (2018–Present).

Languages — Arabic (native); English (professional working proficiency).